

ENVIRONMENTAL SCIENCE AND ENGINEERING

ASSIGNMENT 2: CAMPUS BIODIVERSITY EXPLORER



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1. INTRODUCTION

Biodiversity is the total variety of living organisms in a given area, together with the genetic variation they carry and the ecosystems they build between them. It is usually discussed at the scale of forests and wetlands, yet a built-up institutional campus is itself a working ecosystem. Lawns, avenue trees, ornamental hedges, hostel margins and patches of untended ground each support a distinct assemblage of species, and the interactions between them, such as pollination, herbivory, predation and decomposition, run exactly as they do in wilder habitats.

This assignment documents the biodiversity present within a 500 m radius of the campus of Sri Sivasubramaniya Nadar (SSN) College of Engineering and Shiv Nadar University Chennai, at Kalavakkam on Rajiv Gandhi Salai (OMR), Tamil Nadu. The survey was carried out on foot, and every organism reported here was photographed in the field by the team. No internet images have been used.

Objectives

- To record as many species of plants and animals as possible within the defined study area.
- To identify each organism to species level wherever the photographic evidence allows, and to note the category, location and habitat of each.
- To prepare a systematic biodiversity inventory and an annotated map of observation zones.
- To construct food chains from the recorded organisms and identify the trophic level occupied by each.
- To draw ecological inferences about the condition of the campus habitat and the value of conserving it.

Materials and method

The survey was conducted between approximately 09:00 and 11:00 hours, when reptiles are basking and birds are active. The team walked six pre-assigned zones (Section 2), recording each organism encountered with a smartphone camera. Position was noted against the nearest named campus landmark. Nothing was collected, handled or disturbed at any point: all plants were photographed in situ and all animals from a distance sufficient to avoid disturbing them.

Identification was carried out after the field visit from the photographs, using Google Lens and iNaturalist for a first suggestion and then confirming each name against regional floras and field guides for peninsular India. Where the photograph does not carry enough diagnostic detail for a confident species name, the identification has been deliberately left at genus or order level and flagged as such in the inventory rather than guessed at.

2. STUDY AREA

The campus lies on the western side of Rajiv Gandhi Salai (Old Mahabalipuram Road) at Kalavakkam, roughly 12.7524° N and 80.1953° E, in Chengalpattu District, Tamil Nadu. The site is about 4 km inland from the Bay of Bengal coast and falls in the tropical wet and dry climatic zone, with a mean annual temperature near 28 °C and most of its roughly 1200 mm of rain arriving with the north-east monsoon between October and December. The natural vegetation of this coastal plain is tropical dry evergreen scrub on sandy soils, a formation now almost entirely replaced in the surrounding area by built development.

The land within the survey radius is a mosaic of four habitat types, and this mosaic is what gives the campus its species variety:

- Managed lawn and turf: extensive mown Bermuda grass around the sports complex, cricket field and the Management Block, with low herbaceous weeds persisting between the grass blades.
- Landscaped ornamental beds: irrigated borders and lawn islands planted with palms, cycads and flowering shrubs, which provide a near-continuous nectar supply.
- Avenue and shade trees: mature Neem and Gulmohar lining the internal roads, forming the only substantial canopy layer on site and the main perching and nesting stratum for birds.
- Unmanaged margins and leaf litter: strips along hostel roads and behind buildings where litter accumulates and the soil stays damp, supporting detritivores, spiders and reptiles.

Distribution of survey zones

In line with the instruction to distribute locations between teams, our team surveyed six zones spanning the academic core, the sports precinct, the hostel roads and the auditorium greenbelt. These are marked A to F on the map in Section 4 and referenced in the Observation Zone column of the inventory.

- Zone A — Sports Complex and Cricket Field lawns: open turf and lawn borders.
- Zone B — SSN Canteen Road neem avenue: mature roadside Neem canopy.
- Zone C — Admin and Academic Block surrounds: paved corridors, courtyards, foyer and adjacent avenue trees.
- Zone D — School of Management landscaped garden: irrigated lawn with palms, cycads and flowering hedges.
- Zone E — Gents Hostel Road and Hostel Link Road: shaded road margins with damp soil and leaf litter.
- Zone F — Mini Auditorium and Research Centre Road greenbelt: hedges and shrubbery with weedy undergrowth.

4. CAMPUS BIODIVERSITY MAP

The map below shows the campus with the six observation zones marked A to F, and a dashed circle indicating the 500 m survey radius centred on the academic core. The scale bar and north arrow are given for orientation.

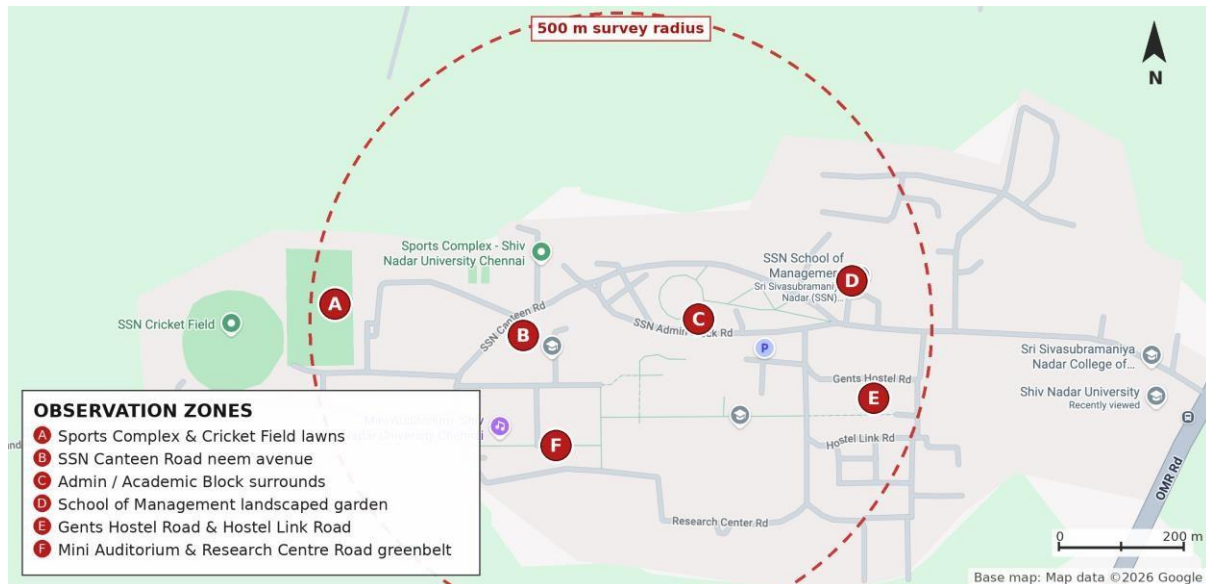


Fig. A. Annotated map of the study area showing observation zones A to F within the 500 m survey radius. Base map: Map data ©2026 Google.

The zones were chosen to sample each of the four habitat types described in Section 2 rather than to cover the area uniformly. Zones A and D represent managed lawn and irrigated ornamental planting, Zone B the avenue canopy, Zone C the built and paved core with its courtyards, and Zones E and F the shaded, littered margins. The clear result of sampling this way is that species richness was not spread evenly across the zones: the two unmanaged margin zones, E and F, produced most of the animal records despite occupying the smallest area, while the extensively landscaped Zone D produced the largest number of plant records but very few animals.

5. FOOD CHAIN

Two food chains have been constructed entirely from organisms photographed during this survey. Both are five links long. The arrows show the direction in which energy flows, that is, from the organism eaten towards the organism eating it.

Food chain 1 — grazing chain of the lawn and avenue

FOOD CHAIN 1 — Campus Lawn & Avenue-Tree Chain



Fig. B. Grazing food chain constructed from species recorded in Zones A, B, C and E.

Neem (*Azadirachta indica*) → Tawny Coster (*Acraea terpsicore*) → Oriental Garden Lizard (*Calotes versicolor*) → Indian Jungle Crow (*Corvus culminatus*) → Indian Pariah Dog (*Canis lupus familiaris*)

The chain begins with the Neem and the lawn herbs, which convert sunlight into chemical energy by photosynthesis and are therefore the producers. The Tawny Coster is the primary consumer: its adults take nectar from the Ixora and Duranta hedges and its caterpillars feed directly on leaf tissue, so energy passes from plant to herbivore. The Oriental Garden Lizard is the secondary consumer, an insectivore that ambushes butterflies, grasshoppers and other insects from the tree bases and litter where it was photographed in Zones E and F. The Indian Jungle Crow is the tertiary consumer, an omnivorous bird that takes small reptiles, nestlings and insects as well as plant matter and refuse. The Indian Pariah Dog closes the chain as a top predator within the study area: no animal on campus preys on an adult dog, and free-ranging dogs are opportunistic enough to take injured birds and small vertebrates as well as scavenge. This last link is the weakest in the chain and is best described as opportunistic predation and scavenging rather than routine predation.

Food chain 2 — detritus chain of the shaded margins

FOOD CHAIN 2 — Detritus (Decomposer) Chain



Fig. C. Detritus food chain constructed from species recorded in Zones B and E.

Neem leaf litter → Millipede (*Anoplodesmus sp.*) → Ground Spider (*Araneae*) → Oriental Garden Lizard (*Calotes versicolor*) → Indian Jungle Crow (*Corvus culminatus*)

This second chain begins not with a living plant but with dead organic matter, the fallen Neem leaves that accumulate along the hostel roads. The millipede is a detritivore, fragmenting and

consuming that litter and in doing so returning nutrients to the soil; it enters the chain at the primary consumer level. The ground spider preys on the millipede and other soil arthropods as a secondary consumer, is itself taken by the garden lizard, and the lizard by the crow. Two features are worth noting: the same two species, the lizard and the crow, occupy the upper levels of both chains, which means the two are not independent but branches of a single food web; and the trophic level of a species is a property of what it eats rather than of the species itself, which is why the lizard sits at level three in chain 1 and level four in chain 2.

Trophic levels of recorded organisms

Trophic level	Species recorded	Ecological role
Producer	Neem, Bermuda Grass, Sessile Joyweed, Ixora	Fix solar energy by photosynthesis; form the base of every chain observed
Primary consumer	Tawny Coster (larva and adult), Millipede	Herbivores and detritivores feeding directly on living leaves or leaf litter
Secondary consumer	Ground Spider, Oriental Garden Lizard	Small carnivores preying on insects and other arthropods
Tertiary consumer	Oriental Garden Lizard, Indian Jungle Crow	Feed on secondary consumers; the lizard occupies both levels depending on prey
Top predator	Indian Jungle Crow, Indian Pariah Dog	No natural predator within the study area; the dog is an opportunistic omnivore

Table 3. Assignment of recorded species to trophic levels.

The shape of both chains reflects the ten per cent law: only about a tenth of the energy at one trophic level is available to the next, the rest being lost as heat in respiration. This is why the campus supports a large biomass of grass and trees, a smaller number of insects, fewer lizards still, and only a handful of crows and dogs. It is also why chains rarely extend beyond four or five links, since there is not enough energy left to sustain a sixth.

6. ECOLOGICAL OBSERVATIONS

Species richness varies with management intensity

The clearest pattern in the data is that animal diversity was highest where human management was lightest. Zones E and F, the shaded and littered road margins, yielded the lizard, the millipede and the spider, that is, most of the animal records, from the smallest surveyed area. The heavily maintained lawn and ornamental beds of Zone D yielded the greatest number of plant species but almost no animals. Frequent mowing removes the flowering weeds and the tall grass that insects need for shelter and egg-laying, and raking away leaf litter removes the substrate that detritivores and the predators dependent on them require. Manicured green space is not the same thing as biodiverse green space.

Ornamental planting sustains pollinators

Against this, the ornamental hedges do real ecological work. The Ixora, Allamanda and Duranta beds flower more or less continuously through the year, and the Tawny Coster in Fig. 14 was photographed moving between the Ixora hedge and the lawn herbs. In a landscape where the original dry evergreen scrub has largely been cleared, these irrigated beds provide a nectar

supply during the dry months when little else is in flower, and so support the pollinator populations on which the remaining native plants also depend.

Litter and moisture drive the decomposer community

The millipede in Fig. 19 was found on damp soil, and millipedes are generally active on the surface only when humidity is high. The spider in Fig. 20 was photographed on bare damp ground, cryptically coloured against it. Both records come from the same shaded strip along Hostel Link Road, and both depend on the leaf litter accumulating there. Decomposers of this kind are the least conspicuous organisms recorded but among the most important, since they release the nutrients locked in dead tissue back into the soil for the producers to take up again.

Synanthropic species dominate the vertebrates

All four vertebrates recorded, the Jungle Crow, the Rock Pigeon, the Garden Lizard and the Pariah Dog, are synanthropic species that thrive alongside human settlement. The pigeon in Fig. 17 was perched on a corridor railing, using the building as it would a cliff face, and the dog in Fig. 18 was resting on a shaded plinth. The absence of any specialist or shy species, and the absence of amphibians, fungi, mosses and lichens from this survey, is informative in itself: it suggests a habitat that is heavily modified and dry at the surface, offering little for species that need undisturbed ground, standing water or high humidity. A survey of the same area during the north-east monsoon would very likely add frogs, fungi and mosses to the list.

Limitations of this survey

- The survey covered a single two-hour morning window in the dry season. Nocturnal species (bats, geckos, moths) and monsoon-dependent groups (amphibians, fungi) were therefore systematically missed.
- Only one visit was made, so the record is a snapshot rather than a measure of abundance; no counts or transect densities were taken.
- Identification rests on photographs alone, which limits confidence for small arthropods and for ornamental cultivars.
- Birds were recorded only where they could be photographed, so several species heard but not photographed are absent from the inventory.

7. PHOTOGRAPHIC DOCUMENTATION

All twenty photographs below were taken by the team during the field visit on 09 August 2026 using a smartphone camera. Each is captioned with the common name, the scientific name in *italics* and the observation zone. No image has been sourced from the internet.



Fig. 1 Yellow Ixora
Ixora coccinea var. *lutea*
Zone D



Fig. 2 Red Ixora
Ixora coccinea
Zone D



Fig. 3 Dwarf Pink Ixora
Ixora cv. 'Dwarf Pink'
Zone F



Fig. 4 Red Ixora in bud, over Diamond Flower
Ixora coccinea / *Oldenlandia corymbosa*
Zone F



Fig. 5 Neem canopy
Azadirachta indica
Zone B



Fig. 6 Indian Jungle Crow in Gulmohar
Corvus culminatus / *Delonix regia*
Zone C



Fig. 7 Betel Nut Palm with fruit bunch
Areca catechu
 Zone D



Fig. 8 Bismarck Palm
Bismarckia nobilis
 Zone D



Fig. 9 Sago Palm
Cycas revoluta
 Zone D



Fig. 10 Ruffled Fan Palms, courtyard
Licuala grandis
 Zone C



Fig. 11 Dumb Cane and Golden Cane Palm
Dieffenbachia seguine / Dyspis lutescens
 Zone C



Fig. 12 Golden Trumpet on lawn border
Allamanda cathartica
 Zone A



Fig. 13 Golden Dewdrop in flower
Duranta erecta
Zone F



Fig. 14 Tawny Coster on Sessile Joyweed
Acraea terpsicore / *Alternanthera sessilis*
Zone A



Fig. 15 Oriental Garden Lizard in leaf litter
Calotes versicolor
Zone E



Fig. 16 Oriental Garden Lizard at tree base
Calotes versicolor
Zone F



Fig. 17 Rock Pigeon on corridor railing
Columba livia
Zone C



Fig. 18 Indian Pariah Dog resting in shade
Canis lupus familiaris
Zone C



Fig. 19 Millipede on damp soil
Anoploidesmus sp.
Zone E



Fig. 20 Ground Spider, cryptic on soil
Order Araneae
Zone E

Figs. 1 to 20. Original photographic record of species observed within the study area.

8. CONCLUSION

This survey recorded twenty-three species within a 500 m radius of the campus, spanning ten major groups from trees and grasses to birds, a butterfly, a reptile, arthropods and a mammal. Two five-link food chains were constructed entirely from organisms photographed on site, one a grazing chain rooted in the Neem and lawn vegetation and the other a detritus chain rooted in leaf litter, and both were found to converge on the same two upper-level consumers, demonstrating that the campus supports not merely isolated chains but an interconnected food web.

The wider finding is that biodiversity on a built campus is concentrated in exactly the places that look least cared for. The unmown margins, the accumulated leaf litter and the shaded damp strips along the hostel roads sustained more animal life than the extensive manicured lawns, while the ornamental hedges proved valuable in a different way, as a year-round nectar source in a landscape stripped of its original scrub vegetation. The absence of amphibians, fungi and lichens marks the limits of the habitat as it currently stands, and also the limits of a single dry-season visit.

Recommendations

- Leave selected margins unmown and allow leaf litter to remain in place beneath avenue trees, to maintain habitat for detritivores and the predators that depend on them.
- Retain and expand native flowering species such as Neem alongside the ornamental beds, so that the nectar supply supports native as well as generalist pollinators.
- Create a small water feature or allow a seasonal depression to hold rainwater, which would very likely bring amphibians and dragonflies onto the campus.
- Repeat the survey during the north-east monsoon to capture the groups this dry-season visit could not record, and maintain a cumulative campus species list across student batches.

Documenting the biodiversity of a familiar place changes how it is seen. Ground that had been walked past daily for years turned out, on close attention, to hold a lizard hunting in the litter, a millipede recycling dead leaves and a butterfly moving between hedges. Conservation on this scale requires no new land, only the willingness to leave some of it alone.

9. REFERENCES

1. Gamble, J. S. and Fischer, C. E. C., Flora of the Presidency of Madras, Adlard and Son, London, 1915 to 1936.
2. Grimmett, R., Inskipp, C. and Inskipp, T., Birds of the Indian Subcontinent, 2nd edition, Oxford University Press, 2011.
3. Kehimkar, I., Butterflies of India, Bombay Natural History Society, Mumbai, 2016.
4. Daniels, R. J. R., Amphibians and Reptiles of Peninsular India, Universities Press, Hyderabad, 2005.
5. iNaturalist, California Academy of Sciences, www.inaturalist.org, accessed August 2026, used for preliminary identification only.
6. India Biodiversity Portal, www.indiabiodiversity.org, accessed August 2026.
7. Base map imagery: Map data ©2026 Google, Google Maps, accessed August 2026.

Declaration: All photographs reproduced in this report were taken by the members of this team during the field visit described. No image has been obtained from the internet or from any other source. No plant or animal was collected, handled, harmed or disturbed in the course of the survey.